

```
class n1 extends object
  field a
  field b

  method initialize(x,y)
    begin
      set a = x;
      set b = y
    end

  method p(x,y)
    send self r(*(a,x),*(b,y))

  method r(x,y)
    +(-(x,y),-(a,b))
```

```
class n2 extends n1
  field a
  field c

  method initialize(x,y)
    begin
      super initialize(-(x,y),+(x,y));
      set a = x;
      set c = y
    end

  method r(x,y)
    +(-(x,y),a,-(b,c))

class n3 extends n2
  field d

  method initialize(x,y)
    begin
      super initialize(+(x,y),-(x,y));
      set d = /(x,y)
    end

  method p(x,y)
    send self r(-(a,x),+(b,y))

  method r(x,y)
    +(+ (x,y),-(a,b),c)
```

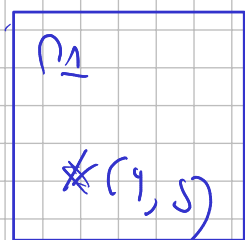
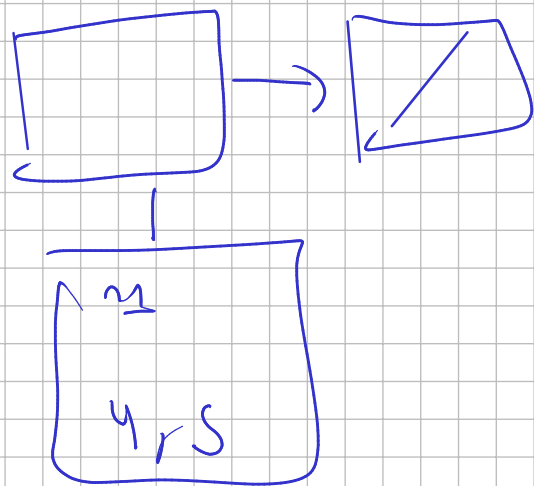
```
let
  → o1 = new n1(4,5)
  o2 = new n2(7,3)
  o3 = new n3(6,7)

  in
    let
      → u = send o1 p(3,5)
      v = send o2 p(3,8)
      z = send o3 p(5,6)

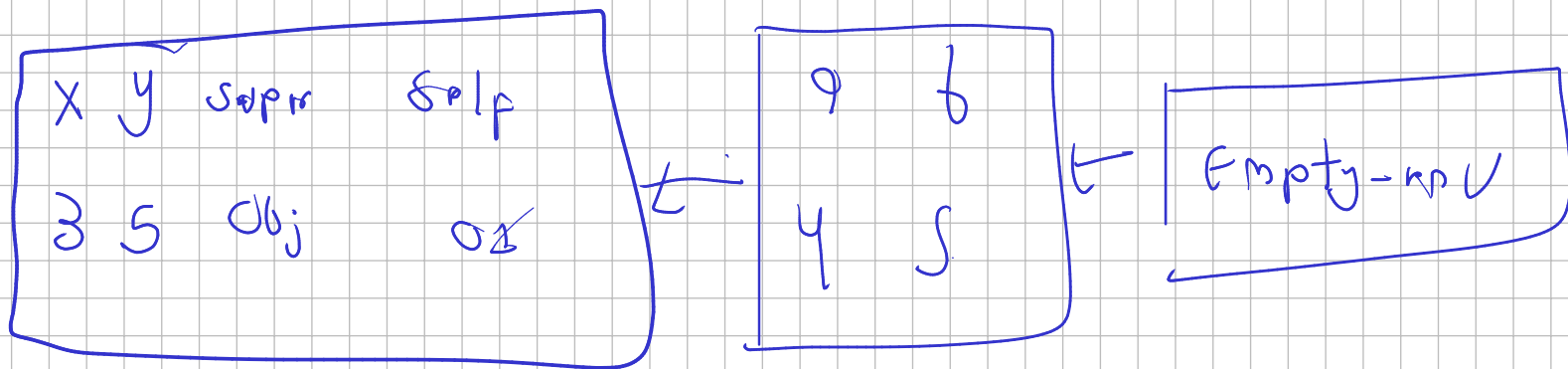
      in
        *(u,v,z)
```



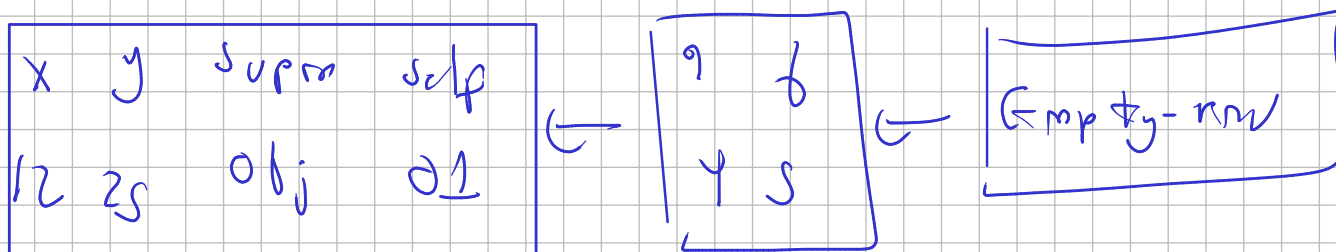
o1)



send o1 p(3, 8)



send o1 r(12, 28)



+ (-13, -1) = -14

```
class n1 extends object
  field a
  field b

  method initialize(x,y)
    begin
      set a = x;
      set b = y
    end

  method p(x,y)
    send self r(*(a,x),*(b,y))

  method r(x,y)
    +(-(x,y),-(a,b))
```

```
class n2 extends n1
  field a
  field c

  method initialize(x,y)
    begin
      super initialize(-(x,y),+(x,y));
      set a = x;
      set c = y
    end

  method r(x,y)
    +(-(x,y),a,-(b,c))

class n3 extends n2
  field d

  method initialize(x,y)
    begin
      super initialize(+(x,y),-(x,y));
      set d = /(x,y)
    end

  method p(x,y)
    send self r(-(a,x),+(b,y))

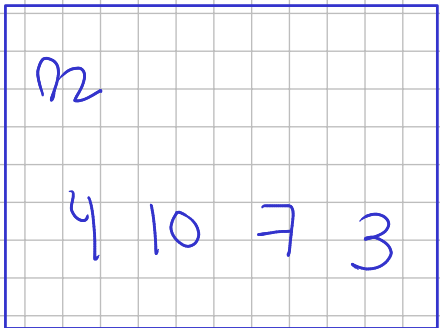
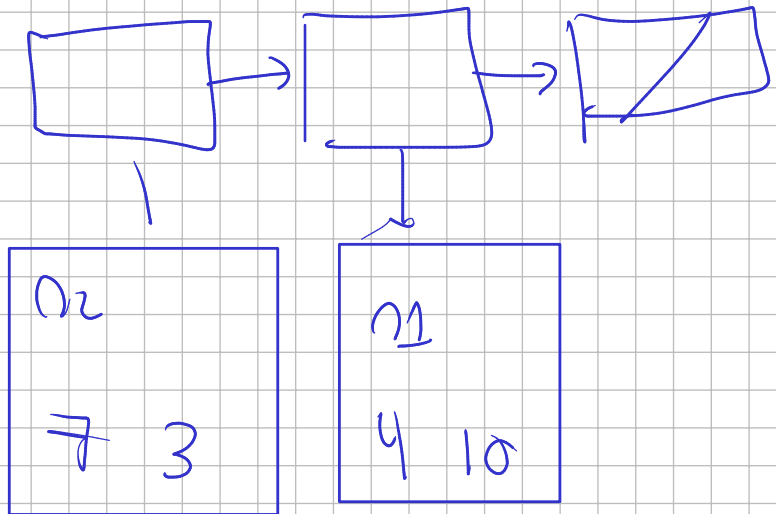
  method r(x,y)
    ++(x,y),-(a,b),c)
```

```
let
  o1 = new n1(4,5)
  o2 = new n2(7,3)
  o3 = new n3(6,7)

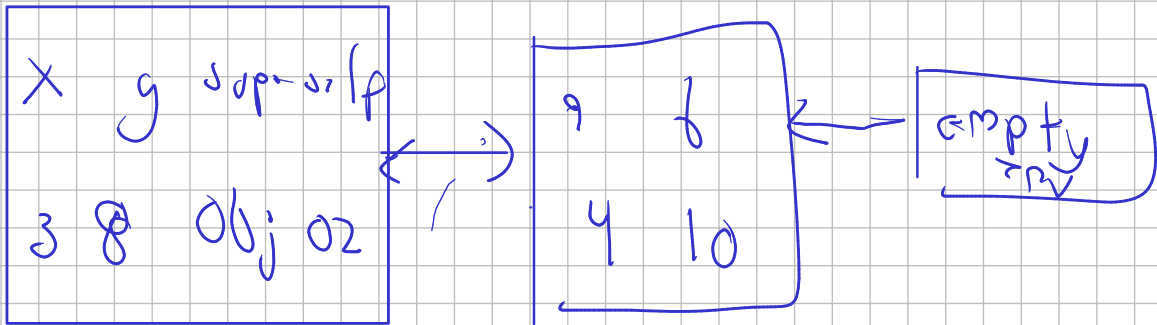
in
  let
    u = send o1 p(3,5)
    v = send o2 p(3,8)
    z = send o3 p(5,6)

  in
    *(u,v,z)
```

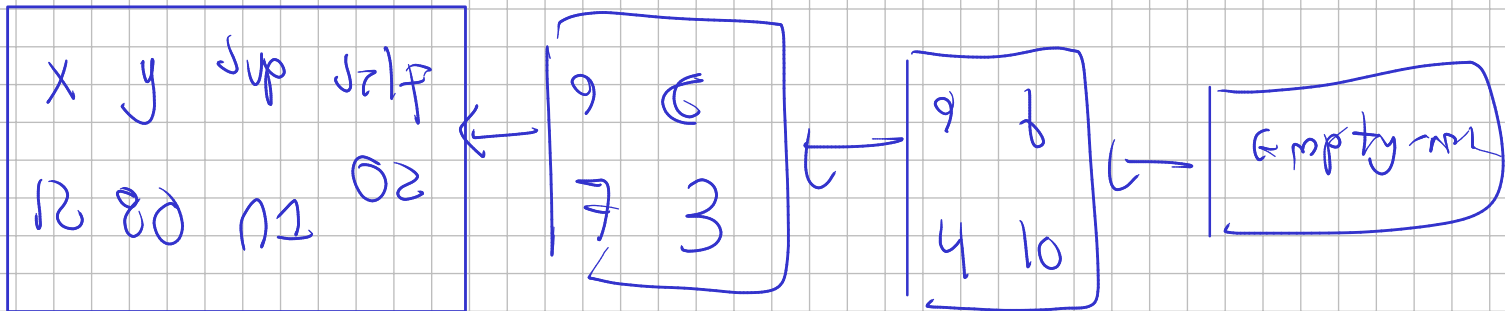
o2



send o2 p(3,8)



send o2 r(12,80)



+(-(x,y),a,-(b,c))
+(-68,7,7)=-54

```
class n1 extends object
  field a
  field b

  method initialize(x,y)
    begin
      set a = x;
      set b = y
    end

  method p(x,y)
    send self r(*(a,x),*(b,y))

  method r(x,y)
    +(-(x,y),-(a,b))
```

```
class n2 extends n1
  field a
  field c

  method initialize(x,y)
    begin
      super initialize(-(x,y),+(x,y));
      set a = x;
      set c = y
    end

  method r(x,y)
    +(-(x,y),a,-(b,c))
```

```
class n3 extends n2
  field d

  method initialize(x,y)
    begin
      super initialize(+(x,y),-(x,y));
      set d = /(x,y)
    end

  method p(x,y)
    send self r(-(a,x),+(b,y))

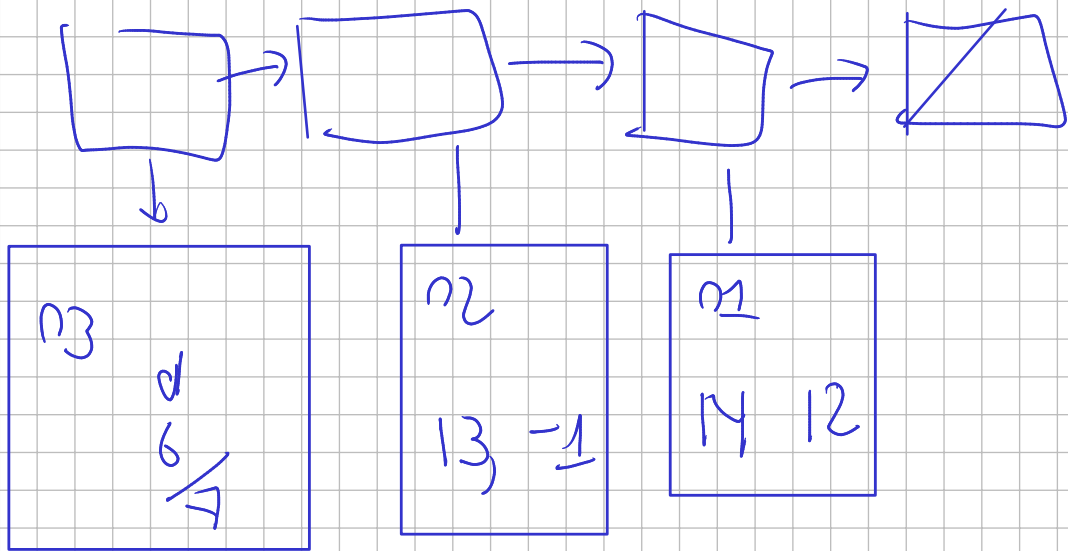
  method r(x,y)
    ++(+(x,y),-(a,b),c)
```

```
let
  o1 = new n1(4,5)
  o2 = new n2(7,3)
  o3 = new n3(6,7)

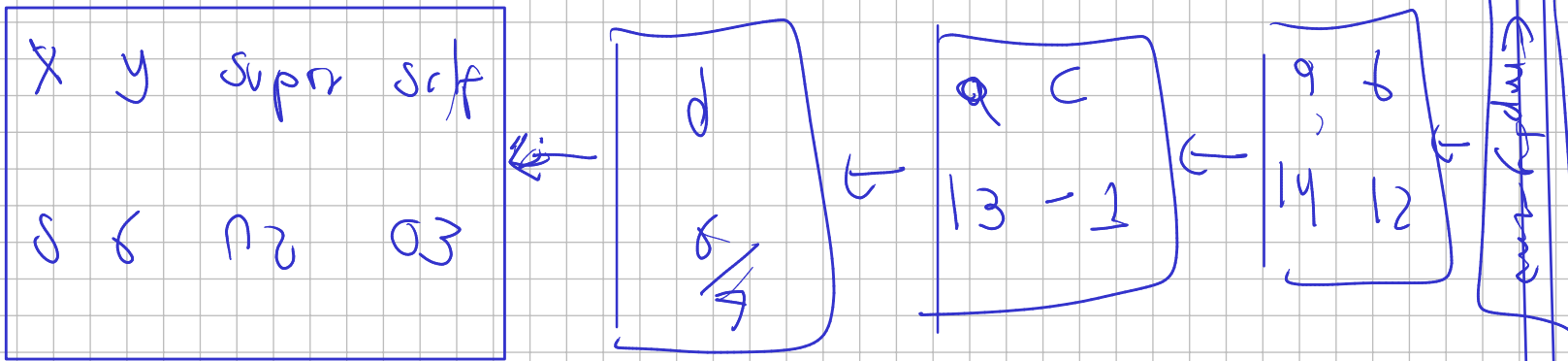
in
  let
    u = send o1 p(3,5)
    v = send o2 p(3,8)
    z = send o3 p(5,6)

  in
    *(u,v,z)
```

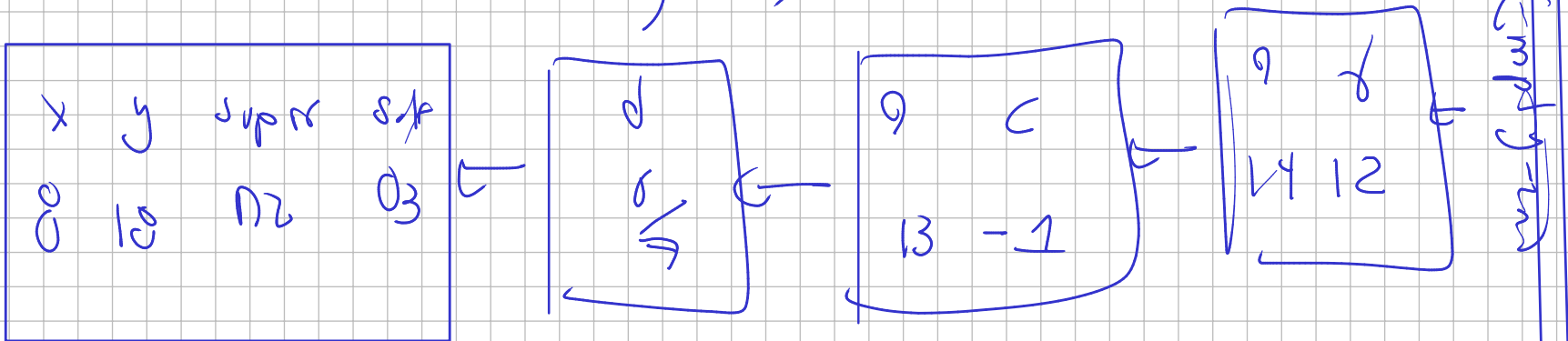
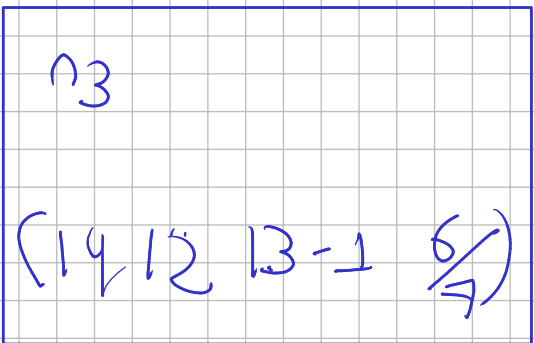
o3



Send o3 p(5,6)



Send o3 r(8,18)



$$+ (+ (8, 18), (13, 12), -1) = + (26, 1, -1) = 26$$